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SIMILARITY MEASURES FOR RETRIEVAL IN CASE-BASED REASONING SYSTEMS

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Case-based reasoning (CBR) is one of the emerging paradigms for designing intelligent systems. Retrieval of similar cases is a primary step in CBR, and the similarity measure plays a very important role in case retrieval. Sometimes CBR systems are called similarity searching systems, the most important characteristic of which is the effectiveness of the similarity measure used to quantify the degree of resemblance between a pair of cases. This article focuses on the similarity measuring methods for CBR and comprises two parts. The first part reviews the existing methods for measuring similarity in the literature based on more than 100 CBR project studies and some general similarity measures seen in other applications. In the second part, a hybrid similarity measure is proposed for comparing cases with a mixture of crisp and fuzzy features. Its application to the domain of failure analysis is illustrated.

Case-based reasoning (CBR) is one of the emerging paradigms for designing intelligent systems. It is an alternative approach to rule-based systems. The advantages of the CBR approach are many, the most important of which is that it parallels the way people solve problems in various situations. It overcomes the drawbacks of rule-based systems with the following advantages:

- CBR does not require an explicit domain knowledge model, and the problem of knowledge elicitation bottleneck can be avoided;
- the main implementation of CBR is reduced to identify significant attributes;
- large volumes of historical cases can be efficiently managed by advanced database techniques; and
- by obtaining new cases, CBR can learn and improve its performance, and maintenance is easier.

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Typically, there are five steps in CBR: presentation of a new case, retrieval of the most similar cases from the case base, adaptation of the selected cases, validation of the current solution given by CBR, and updating of the system by adding the verified solution to the case base. The corresponding methodology for developing a CBR system should closely follow these steps. Among them, case retrieval is a primary step in CBR applications. Without effective case retrieval, a CBR system cannot do anything. Of course, enough cases should be first saved in the case memory, so that a similar one can be retrieved.

The similarity measure used to quantify the degree of resemblance between a pair of cases plays a very important role in case retrieval. Therefore CBR systems are sometimes called similarity searching systems. Because of the importance of similarity measure in CBR, this article focuses on the similarity measuring methods. The article is divided into two parts. Part 1 reviews the existing methods for measuring similarity in the open literature based on more than 100 CBR project studies and some general similarity measures seen in other applications including those using crisp and fuzzy set theories. In part 2, a hybrid similarity measure is proposed to compare cases having a mixture of crisp and fuzzy features. Its application to the domain of failure analysis is demonstrated. The conclusions are given in the final section.

SIMILARITY MEASURES FOR CBR

There are two major case retrieval approaches:

- One is the distance-based or computational approach, which calculates the distance between cases from a number of objects constituting the cases. The most similar case is determined by the evaluation of a similarity measure.
- Second is the indexing or representational approach. The similar case is coded into the structure of the case base itself. The cases are connected by indexing structures. The indexing structure can be traversed to search the similar case. Another situation is based on the graphical appearance of the cases, which compares abstracted bitmaps of design pictures.

Note that the above two approaches might be used in combination. Our similarity measures discussed below will focus on the first approach because it is the most widely used in CBR applications.

Basic CBR Similarity Measures

Commonly, CBR systems use the inverse of weighted normalized Euclidian distance or Hamming distance as the similarity measure. That is,

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = 1 - \mathrm{DIST}(\mathbf{X}, \mathbf{Y}) = 1 - \sqrt{\sum_i w_i^2 \, \mathrm{dist}^2(x_i, y_i)} \tag{1a}$$

use Euclidian distance, and

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = 1 - \mathrm{DIST}(\mathbf{X}, \mathbf{Y}) = 1 - \sum_i w_i \, \mathrm{dist}(x_i, y_i) \tag{1b}$$

use Hamming distance, where weight w_i is normalized for denoting the importance of the ith attribute and $i = 1, 2, \ldots, n$, n being the number of attributes in the cases. The normalized $\mathrm{dist}(x_i, y_i)$ is often represented as

$$\mathrm{dist}(x_i y_i) = |x_i - y_i| / |\max_i - \min_i|$$

For numerical attributes, $\max_i$ and $\min_i$ are the maximum and minimum values of the ith attributes, respectively. For symbolic attributes, if $x_i = y_i$, $\mathrm{dist}(x_i, y_i) = 0$; otherwise, $\mathrm{dist}(x_i, y_i) = 1$. Equation (1) shows that when DIST = 0, SIM attains the maximum value 1, and this means two cases are identical. When DIST = 1, the SIM has a minimum value 0, meaning two cases are completely different.

Another widely used similarity measure is based on the ratio mode proposed by Tversky (1977):

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \frac{\alpha \times \mathrm{common}}{\alpha \times \mathrm{common} + \beta \times \mathrm{different}} \tag{2}$$

where *common* (*different*) represents the number of attributes whose values are classified as similar (dissimilar) between the old case and the new case. Two values are classified as similar (dissimilar) if their similarity is above (below) some threshold; α and β are the corresponding weights for *common* and *different*.

Hunt et al. (1995) proposed another similarity measure based on metaphors taken from the human immune system. The measure assumes the new case to be a new antigen and the old case to be an old antibody. The match score is calculated as

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \mathrm{count} + \sum_i 2^{l_i} \tag{3}$$

where *count* is the number of fields that match between the antigen and the antibody, and l_i is the length of each region consisting of two or more matches. In this measure, case attributes are coded as strings in some specific

order. For example, two cases **X**, **Y** have matching score 13 as illustrated below:

new case (**X**):	Adventure	2498	2	Egypt	Car	14	March	*
old case (**Y**):	Adventure	3066	3	Egypt	Car	14	April	*
count	1	0	0	1	1	1	0	$1 \Rightarrow 5$

SIM(**X**, **Y**): $5 + 2^3 \Rightarrow 13$

Sebag and Schoenauer (1993) discussed a rule-based similarity measure. Let Ω be the problem domain. Given two examples (cases) **X** and **Y** in Ω, the similarity between them SIM(**X**, **Y**) was considered to be the number of rules that are fired by both or by none of **X** and **Y**. Each rule R_i learned from the case base is assigned a weight $w(R_i)$ depending upon the number of training examples it covers. Thus the similarity measure is defined as

$$\forall \mathbf{X}, \mathbf{Y} \in \Omega \qquad \mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \sum_i w(R_i). \tag{4}$$

Because case matching is not a symmetrical operation in some situations, Weber (1994) used the contrast model (Tversky, 1977), in which the similarity between a target case **X** (new case) and a base case **Y** (old case) was given by the following formula:

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \theta f(\mathbf{X} \cap \mathbf{Y}) - \alpha f(\mathbf{X} - \mathbf{Y}) - \beta f(\mathbf{Y} - \mathbf{X}) \tag{5}$$

where the intersection $(\mathbf{X} \cap \mathbf{Y})$ describes the set of attributes that are common to **X** and **Y**, and the complement sets (**X**-**Y**) and (**Y**-**X**) describe attributes observed only in the target case and attributes observed only in the base case, respectively. Here θ, α, β are the weighting parameters denoting different importance for each term. The f is some operator or algorithm to compute the matching score of the related sets. Pal and Campbell (1995) defined the mutual similarity coefficient for **X** and **Y** as

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = (S_{lr} + S_{rl})/2 \tag{6}$$

where S_{lr} is the normalized association of **X** with respect to **Y** and S_{rl} is the normalized association of **Y** with respect to **X**. S_{lr} is defined as $|\mathbf{X} \cap \mathbf{Y}|/|\mathbf{X}|$ denoting the ratio of the number of attributes common to **X** and **Y** to the size of **X**. Similarly, S_{rl} is defined by replacing $|\mathbf{X}|$ with $|\mathbf{Y}|$.

Stanfill and Waltz (1986) computed the distance DIST between two cases **X** and **Y** as

$$\mathrm{DIST}(\mathbf{X}, \mathbf{Y}) = \sum_i w_i(x_i) \times \mathrm{diff}(x_i, y_i) \tag{7}$$

The similarity between individual (symbolic) feature values was calculated using a value-difference-metric (VDM). The weight of the ith attribute with value x_i, $w_i(x_i)$, and the difference between two attribute values, $\text{diff}(x_i, y_i)$, are defined as follows:

$$w_i(x_i) = \sqrt{\sum_{c \in C} \left[\lambda(c, x_i) / \lambda(x_i) \right]^2} \tag{8}$$

$$\text{diff}(x_i, y_i) = \sum_{c \in C} \left(\frac{\lambda(c, x_i)}{\lambda(x_i)} - \frac{\lambda(c, y_i)}{\lambda(y_i)} \right)^2 \tag{9}$$

where $i = 1, \ldots, n$, n being the number of attributes, $\lambda(c, x_i)$ is the number of times that cases of class c in the case library have value x_i for the ith attribute, whereas $\lambda(x_i)$ is the same value summed over the set of all classes C. VDM assigns higher weights to attributes whose distribution of values across classes is highly skewed and greater differences to values whose corresponding sets of cases have highly disparate class distributions. Thus two cases are similar if they have attribute values whose respective projections on the training library have similar class distributions.

Enhancement of Basic CBR Similarity Measures

Case Typicality

CBR used for supporting human decision such as classification and planning implicitly expresses the knowledge in cases. It is natural that some cases contain more knowledge than others. The former are more typical cases in the case space. Agre (1995) proposed a method that considers case typicality, that is,

$$\text{DIST}(\mathbf{X}, \mathbf{Y}) = W_{\mathbf{Y}} \times \text{Dist}(\mathbf{X}, \mathbf{Y}) \tag{10}$$

where $\text{Dist}(\mathbf{X}, \mathbf{Y})$ is the distance between new case $\mathbf{X}$ and old case $\mathbf{Y}$ without considering the case typicality. $W_{\mathbf{Y}}$ is the weight of old case $\mathbf{Y}$. The weight of an old case is simply the reciprocal of its typicality. When more than one case belonging to different classes is found, the most typical one is preferred. The case typicality can be determined by the expert or by calculating the average distance between the tested case and selected typical cases.

Attribute Relevance

To reflect the attribute relevance, Cain et al. (1991) proposed an approach based on case explanation. They used a parameterized similarity function influenced by the explanations produced by the domain theory for

each case in memory:

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \frac{\alpha \sum_i \mathrm{sim}(x_i, y_i) + \beta \sum_i \mathrm{relevance}(y_i) \times \mathrm{sim}(x_i, y_i)}{\alpha n + \beta \sum_i \mathrm{relevance}(y_i)} \tag{11}$$

where

$$\begin{aligned} \mathrm{sim}(x_i, y_i) &= 1 \qquad x_i = y_i \\ \mathrm{sim}(x_i, y_i) &= 0 \qquad x_i \neq y_i \\ \\ \mathrm{relevance}(y_i) &= 1 \qquad y_i \text{ is relevant} \\ \mathrm{relevance}(y_i) &= 0 \qquad y_i \text{ is irrelevant} \end{aligned}$$

Here y_i is a fact in the past problem and x_i is a fact in the new problem. A fact y_i is relevant if it is a premise for an explanation in the past case; α, β are the corresponding term weights. When β is set to zero, the measure ignores the relevance of facts, and a pure similarity-based match is performed.

Transforming Attributes

Different problems usually have different attributes whose expressing methods and measuring scales may be quite different. It is sometimes necessary to transform original problems and their attributes into equivalent ones in order to apply the CBR method to help in problem solving. To illustrate the equivalence of the learning power of symbolic and case-based methods, Wess and Globig (1993) studied the transformation of a simple symbolic learning algorithm (the version space algorithm) into an equivalent case-based variant. Symbolic learning approaches represent the concepts learned explicitly, while case-based approaches describe concepts implicitly by a pair (CB, SIM), i.e., by a measure of similarity SIM and a set of cases CB. In the version space algorithm, the main learning task is to distinguish between relevant and irrelevant attributes. An attribute value is called relevant if it is part of the target concept $\mathbf{X} = (x_1, \ldots, x_n)$. Function f_i is defined as

$$\begin{aligned} f_i(y_i) &= 1 \qquad x_i = y_i \\ f_i(y_i) &= 0 \qquad \text{otherwise} \end{aligned}$$

The functions f_i are then combined into one $f\colon U \rightarrow \{0, 1\}^n$, $f[(x_1, \ldots, x_n)] = [f_1(x_1), \ldots, f_n(x_n)]$. The distance between two cases $\mathbf{X}$ and $\mathbf{Y}$ is defined using the city-block metric as follows:

$$\mathrm{DIST}(\mathbf{X}, \mathbf{Y}) = |f_1(x_1) - f_1(y_1)| + \cdots + |f_n(x_n) - f_n(y_n)| \tag{12}$$

Every change of the $f_i (i = 1, \ldots, n)$ causes a change of the underlying measure DIST. Case-based learning may be seen as a concept-forming task. For each intended concept, the function f_i is learned by some algorithm through selected positive and negative cases. Once the concept is learned, the function f_i and the CB can be used for classification.

Adaptability Guided Measure

When a CBR system is measured with a new case, it will retrieve the most similar case from its case base and, if necessary, adapt it to provide the desired solution. Smyth and Keane (1993) argued that the success of CBR is critically dependent upon the retrieval of a suitable case that is the easiest, of those available, to adapt. Assume that the case attributes are {Action, Vehicle, Direction, Destination, Speed}. Case **Y1** is {Move, Buggy, Forward, Tension-Reel, 1-speed}. Case **Y2** is {Lift, Lifter, Upward, Spool-stand, 2-speed}. Target case **X** is {Move, Buggy, Forward, Tension-Reel, 2-speed}. Measuring the semantic similarity between the target and candidates, case **Y1** is retrieved. However, according to the domain knowledge, modifying the speed of a case is far more difficult than modifying the other attributes. So, although the similarity between the target and case **Y1** is higher, case **Y2** is easier to adapt. One way to consider attribute adaptability is to assign an appropriate weight to the corresponding attribute. Smyth and Keane used a measure of the closeness between the target and candidates in the adaptation space. The retrieval space and adaptation space are linked using the adaptation knowledge. How to measure the attribute adaptability is a major issue in applying this concept.

Missing Attributes

In reality, the stored case and the target case will commonly contain some missing or null attribute values. These missing attribute values might appear in both cases or just one. Handling missing values is an important issue in similarity measure. Agre (1995) used the following method to estimate the distance of an attribute with missing values:

$$\text{dist}^2(x_i, y_i) = 1/L_i \times (1 - 1/L_i) \tag{13}$$

where L_i is the number of possible values of the ith attribute that is restricted to the symbolic type. Ricci and Avesani (1995) computed the similarity between two cases **X** and **Y** for normalized attribute i with missing value as follows:

$$\text{sim}(x_i, y_i) = 0.5 \qquad \text{if } x_i \text{ or } y_i \text{ is unknown} \tag{14}$$

Surma and Vanhoof (1995) estimated the distance of missing values differently. For both unknown values, dist(x_i, y_i) = 0. If one attribute value is unknown and the other is known, then their distance is 1. Here attribute value is also normalized.

Summary

The similarity measures and related considerations discussed above are typical in the CBR literature. Some more complex measures are excluded because they are more or less context dependent. Another important way to improve measure performance is by weighting attributes according to their relative importance. Equation (8) describes one method to assign attribute weights. The method belongs to the IGNORANT category. Wettschereck and Aha (1995) compared five weighting methods. Overall, the basic principles for designing similarity measures are effectiveness and simplicity. The balance between them depends on the application.

OTHER SIMILARITY MEASURES

The notion of similarity plays a fundamental role in the knowledge and behavior theories. Besides the measures discussed in the previous section, the following general similarity measures can be potentially adopted for CBR applications.

Similarity Measure for Crisp Sets

Assume **X** and **Y** are crisp sets in n-dimensional space, $\mathbf{X} = (x_1, x_2, x_3, \ldots, x_n)$, $\mathbf{Y} = (y_1, y_2, y_3, \ldots, y_n)$, where x_i and y_i can only take 1 or 0. If x_i (or y_i) appears in **X** (or **Y**), it takes 1; otherwise, it takes 0. A list of typical similarity measures for crisp sets are as follows:

Inner product

$$|\mathbf{X} \cap \mathbf{Y}| \tag{15}$$

Dice coefficient

$$\frac{2|\mathbf{X} \cap \mathbf{Y}|}{|\mathbf{X}| + |\mathbf{Y}|} \tag{16}$$

Cosine coefficient

$$\frac{|\mathbf{X} \cap \mathbf{Y}|}{\sqrt{|\mathbf{X}|} \times \sqrt{|\mathbf{Y}|}} \tag{17}$$

Jaccard coefficient

$$\frac{|\mathbf{X} \cap \mathbf{Y}|}{|\mathbf{X}| + |\mathbf{Y}| - |\mathbf{X} \cap \mathbf{Y}|} \tag{18}$$

Overlap coefficient

$$\frac{|\mathbf{X} \cap \mathbf{Y}|}{\min(|\mathbf{X}|, |\mathbf{Y}|)} \tag{19}$$

where $|\mathbf{X}|$ is the number of elements in $\mathbf{X}$, $|\mathbf{Y}|$ is the number of elements in $\mathbf{Y}$, and $|\mathbf{X} \cap \mathbf{Y}|$ is the number of elements appearing jointly in $\mathbf{X}$ and $\mathbf{Y}$.

The inner product is, in principle, unbounded, while the other four coefficients are normalized. Other methods to measure the similarity between two crisp sets with numeric values are transformed measures such as

$$\mathrm{SIM}(\mathbf{X}, \mathbf{Y}) = \frac{2 \sum_i x_i y_i}{\sum x_i^2 + \sum y_i^2} \tag{20}$$

Similarity Measure for Fuzzy Sets

Since the first publication of fuzzy set theory by Zadeh (1965), great progress has been made in this area during the last three decades. One of the major issues is the similarity measure between two fuzzy sets (attribute values) (e.g., Zwick et al., 1987 ; Dubois et al., 1988 ; Miyamoto, 1990a, b ; Gu & Dubuisson, 1990 ; Pappis & Karacapilidis, 1993 ; Hyung et al., 1994 ; Chen et al., 1995 ; Wang et al., 1995 ; Liao & Zhang, 1996). The fuzzy similarity measures are usually based on the geometric model, the set-theoretic approach, and so on.

Measure Based on the Union and Intersection Operations

Assume $\mathbf{A}$ and $\mathbf{B}$ are fuzzy sets ; $\mathbf{a}$ and $\mathbf{b}$ are the vector representations of the fuzzy sets $\mathbf{A}$ and $\mathbf{B}$, respectively. The degree of similarity $M1_{A,B}$ is defined as follows:

$$M1_{A,B} = \frac{\sum_i (a_i \wedge b_i)}{\sum_i (a_i \vee b_i)} \tag{21}$$

where $\wedge$ ($\vee$) represents the min (max) operator, $\mathbf{a} = (a_1, a_2, \ldots, a_n)$, and $\mathbf{b} = (b_1, b_2, \ldots, b_n)$, $\mathbf{a} \in \mathbf{A}$, $\mathbf{b} \in \mathbf{B}$, n is the number of discrete values in the universe of discourse. $\mathbf{A}$ and $\mathbf{B}$ are considered to be approximately equal

(denoted as $\mathbf{A} \sim \mathbf{B}$) if, for a given small nonnegative number ε, $M1_{A,B} \geq \varepsilon$. The number ε is said to be the proximity measure of **A** and **B**. This definition also applies to $M2_{A,B} \sim M6_{A,B}$ discussed below.

Measure Based on the Maximum Difference

The degree of similarity $M2_{A,B}$ between fuzzy sets **A** and **B** is defined as follows:

$$M2_{A,B} = 1 - \max_i(|a_i - b_i|) \tag{22}$$

Measure Based on the Difference and the Sum of Grades of Membership

The degree of similarity $M3_{A,B}$ between a pair of fuzzy sets **A** and **B** is calculated as

$$M3_{A,B} = 1 - \frac{\sum_i |a_i - b_i|}{\sum_i (a_i + b_i)} \tag{23}$$

or

$$M3_{A,B} = 1 - \frac{\sum_i |a_i \vee b_i - a_i \wedge b_i|}{\sum_i (a_i + b_i)} \tag{24}$$

Measure Based on the Geometric Distance Model

Consider that **a** and **b** are two points in n-dimensional space with components (a_i, b_i), $i = 1, 2, \ldots, n$, the degree of similarity $M4_{A,B}$ is defined based on the Minkowski r-metric ($r = 1$) as follows:

$$M4_{A,B} = 1 - \frac{\sum_i |a_i - b_i|}{n} \tag{25}$$

Measure Based on the Set-Theoretic Approach

Assume **A** and **B** are fuzzy sets in the universe of discourse **U**, the degree of similarity $M5_{A,B}$ between **A** and **B** is defined as follows:

$$M5_{A,B} = \sup_{x \in U} \mu_{A \cap B}(x) = \max_{x \in U} \min(\mu_A(x), \mu_B(x)) \tag{26}$$

where $\mu_A(x)$, $\mu_B(x)$, and $\mu_{A \cap B}(x)$ are the membership functions of **A**, **B** and the intersect of **A** and **B**, respectively. Note that when $\mathbf{A} = \mathbf{B}$, $M5_{A,B}$ does not always have value 1.

Measure Based on the Matching Function

In this method, the degree of similarity between two fuzzy sets **A** and **B** is defined as follows:

$$M6_{A,B} = \frac{\mathbf{a} \cdot \mathbf{b}}{\max(\mathbf{a} \cdot \mathbf{a}, \mathbf{b} \cdot \mathbf{b})} \tag{27}$$

Other Similarity Measures

Wang et al. (1995) investigated another similarity measure inspired by the equivalence

$$\mathbf{A} = \mathbf{B} \Leftrightarrow \mathbf{A} \subseteq \mathbf{B} \cap \mathbf{B} \subseteq \mathbf{A}$$

This measure is defined as

$$E_{\wp}(\mathbf{A}, \mathbf{B}) = \min((\inf_{x \in U} \wp(\mathbf{A}(x), \mathbf{B}(x)), \inf_{x \in U} \wp(\mathbf{B}(x), \mathbf{A}(x))) \tag{28}$$

where $\wp$ is an implication operator, i.e., a $[0,1]^2 \rightarrow [0,1]$ mapping satisfying $\wp(0, 0) = \wp(0, 1) = \wp(1, 1) = 1$ and $\wp(1, 0) = 0$. The implication operator can be one of the following:

1. Gödel implication operator

$$\wp(x, y) = 1 \quad x \leq y$$
$$\wp(x, y) = y \quad \text{otherwise}$$

2. Goguen implication operator

$$\wp(x, y) = 1 \qquad x \leq y$$
$$\wp(x, y) = y/x \qquad \text{otherwise}$$

3. Luckasiewice implication operator

$$\wp(x, y) = \min(1, 1 - x + y)$$

Table 1 compares some similarity measures discussed above and summarizes their main properties. In the table, $\mathbf{A}^{\wedge}$ denotes the complement of $\mathbf{A}$,

Table 1 Main Properties of Fuzzy Similarity Measures

Properties	$M1_{A,B}$	$M2_{A,B}$	$M3_{A,B}$	$M4_{A,B}$	$M5_{A,B}$	$M6_{A,B}$
$M_{A,B} = M_{B,A}$	Y	Y	Y	Y	Y	Y
$\mathbf{A} = \mathbf{B} \Leftrightarrow M_{A,B} = 1$	Y	Y	Y	Y	N	Y
$\mathbf{A} \cap \mathbf{B} = 0 \Leftrightarrow M_{A,B} = 0$	Y	N	Y	N	Y	Y
$M_{A,A^{\wedge}} = 1 \Leftrightarrow \mathbf{A} = \mathbf{H}$	Y	Y	Y	Y	N	Y
$M_{A,A^{\wedge}} = 0 \Leftrightarrow \mathbf{A} = \mathrm{I}$ or $\mathbf{A} = 0$	Y	N	Y	Y	N	Y
$\mathbf{A} \sim \mathbf{B} \Rightarrow \mathbf{A} \cup \mathbf{C} \sim \mathbf{B} \cup \mathbf{C}$	N	Y	Y	N	N	N
$\mathbf{A} \sim \mathbf{B} \Rightarrow \mathbf{A} \cap \mathbf{C} \sim \mathbf{B} \cap \mathbf{C}$	N	Y	N	N	Y	Y
$\mathbf{A} \sim \mathbf{B} \Rightarrow \mathbf{A} \circ \mathbf{R} \sim \mathbf{B} \circ \mathbf{R}$	N	Y	N	N	Y	N
$\mathbf{R} \sim \mathbf{S} \Rightarrow \mathbf{A} \circ \mathbf{R} \sim \mathbf{A} \circ \mathbf{S}$	N	Y	N	N	N	N

superscript zero denotes a max-min composition, **H** denotes 0.5 fuzzy set, **R** and **S** are any fuzzy sets, and $\cap$ and $\cup$ denote the intersection and union of two fuzzy sets, respectively.

HYBRID SIMILARITY MEASURE FOR FAILURE ANALYSIS

Zwick et al. (1987) and Liao and Zhang (1996) investigated the fuzzy similarity measures in various fields such as fuzzy expert system, fuzzy pattern recognition, and fuzzy query. Other authors (e.g., Willett, 1987 ; Salton, 1989 ; Miyamoto, 1990a, b ; Cain et al., 1991 ; Artymiuk et al., 1992 ; Gesu, 1994 ; Hyung et al., 1994 ; Kraslawski et al., 1995) discussed various applications of similarity measures. Because most of the traditional tools for formal modeling, reasoning, and computing are crisp, deterministic, and precise in nature, it is no surprise that the majority of proposed similarity measures in the CBR field are based on binary logic or crisp set, that is, the Yes-or-No type rather than the More-or-Less type. Kolodner (1993) and Watson and Marir (1994) stressed the importance of similarity function in CBR but did not discuss fuzzy similarity measure at all. After a careful survey, it was found that only Lee et al. (1995) used fuzzy set theory in CBR. However, linguistic attributes were not considered.

It has been recognized that fuzzy set theory can model reality more naturally and adequately. The fact that CBR systems used to handle human judgment, evaluation, reasoning, and decision cannot handle fuzzy concepts is a major constraint. To make the CBR system more powerful, the incorporation of fuzzy set theory is needed. In the following, the domain of failure analysis (Esaklul, 1990) is used to illustrate that applications with a mixture of crisp and fuzzy features are common in the real word. Subsequently, a hybrid similarity measure is proposed.

Characteristics of Failure Cases

Our failure cases were obtained from Scientific Testing Lab. Inc., a Louisiana company that has been conducting failure analysis business for more than 20 years. A CBR system is being developed to store failure analysis cases and to select most similar cases from the stored case base.

The data acquisition in failure analysis is an incremental process. During different stages of analysis, different actions and tests (e.g., chemical analyses, mechanical tests) are performed to identify the cause of a failure. Unknown or missing attribute values are filled in as the analysis progresses. Depending upon the failure, the total number of required tests vary. The CBR system might be run at different stages of the analysis.

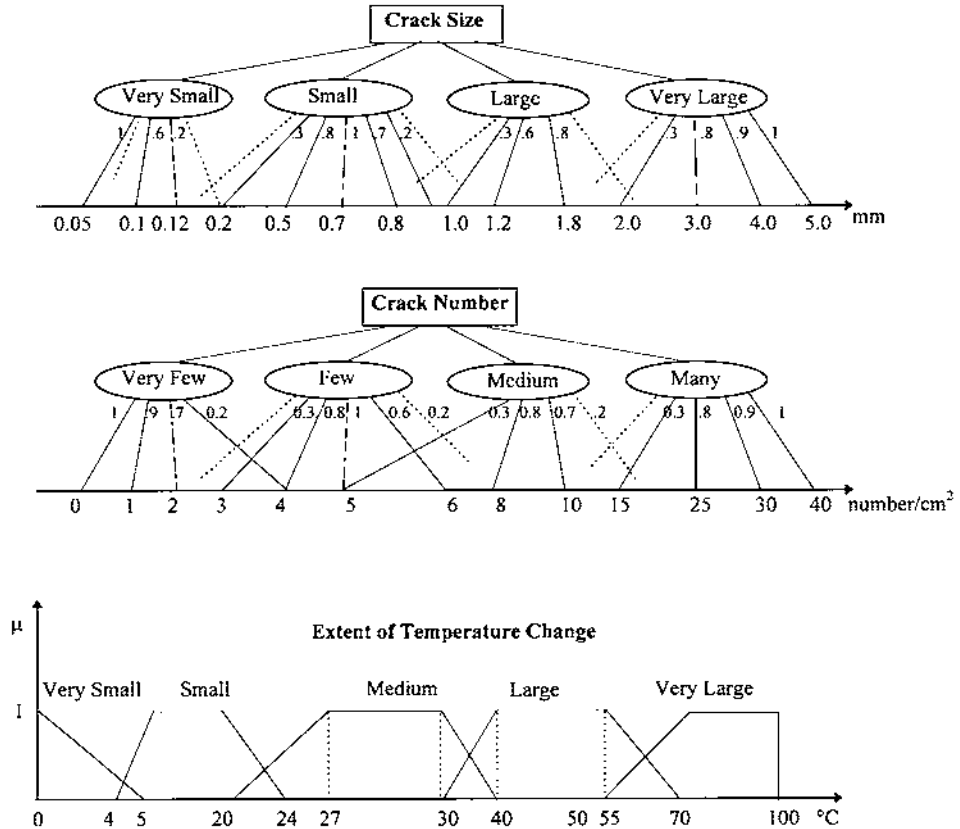


FIGURE 1. Examples of linguistic attributes.

The attributes used to represent failure cases could be associated with the following values:

- Crisp numeric values can be continuous or discrete. For example, the value of attribute "time in service" could be "5.5 years" or "4 years," depending upon the accuracy required.
- Crisp symbolic values are expressed as terms with definite meanings. For example, the value of the attribute "component form" is either "plate" or "pipe," etc. For improving user interactions and failure analysis process, a vocabulary of precise and consistent symbolic representation is needed.
- Fuzzy linguistic values can be considered as linguistic variables. The values of this kind of attribute form a term set. Each term in the term set corresponds to a fuzzy concept such as in Figure 1. Their definitions depend heavily on the application.

- Fuzzy numbers or fuzzy intervals refer to an inexact number of interval with inexact bounds. For example, the pressure is about 220 kPa or the pressure ranges roughly between 100 and 150 kPa as shown in Figure 2.

Other characteristics involved in failure cases include the following:

- Attribute relevance: each case attribute contributes differently to the case matching process. The same attribute in different analysis stages might be of different relevance.
- Missing attributes: in the beginning of the analysis, the values of many attributes of the target case are unknown. Their values might not be known eventually if these attributes are considered not needed for the analysis. Missing attribute values are also generally present in the stored cases.
- Dominant attributes: some attributes are more important than others. This kind of attribute should be adequately weighted.
- Case stability: although each failure case might be unique, the causal relations for failures are relatively stable. In other words, the numbers of potential attributes contributing to each failure mode remain very much the same.

Hybrid Similarity Measure

In developing CBR systems for failure analysis or similar applications, how to measure fuzzy attributes is a major issue in designing a similarity measure. There are two issues involved: the similarity of two fuzzy concepts and the aggregation of all fuzzy attributes. The conventional similarity measures for CBR reviewed in the section above, Similarity Measures for CBR, do not provide a suitable solution for this problem.

Before applying fuzzy similarity measures, the membership functions of fuzzy concepts need to be defined. An LR representation of fuzzy sets (DuBois and Prader, 1980) is used in our study. Usually, fuzzy concepts or sets can be described by Gaussian membership functions, but the computation of Gaussian functions is complex. To make the computation easier,

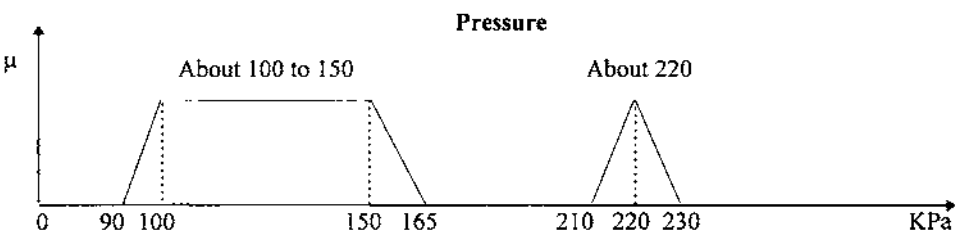


FIGURE 2. Examples of fuzzy number and interval attributes.

trapezoidal fuzzy sets are adopted to model the fuzzy attributes. The shape functions are as follows:

$$L(x) = R(x) = \max(0, 1 - x)$$

The membership function of the fuzzy set $\tilde{M}$ with parameter $\underline{m}$, $\bar{m}$, p, and q is

$$\mu_{\tilde{M}}(x) = \begin{cases} L\left(\dfrac{\underline{m} - x}{p}\right) & x \le \underline{m} \\ 1 & \underline{m} \le x \le \bar{m} \\ R\left(\dfrac{x - \bar{m}}{q}\right) & x \ge \bar{m} \end{cases}$$

For triangular fuzzy sets, $\underline{m} = \bar{m}$. The spreads p and q depend on the attributes. Taking the "extent of temperature change" attribute as an example of fuzzy linguistic attributes, the membership functions for "medium" and "large" are defined as follows:

$$\begin{aligned} \mu_{\tilde{M}}(x) &= L((27 - x)/7) && x \le 27 \\ \mu_{\tilde{M}}(x) &= 1 && 27 \le x \le 30 \\ \mu_{\tilde{M}}(x) &= R((x - 30)/10) && x \ge 30 \end{aligned} \tag{29}$$

$$\begin{aligned} \mu_{\tilde{L}}(x) &= L((40 - x)/10) && x \le 40 \\ \mu_{\tilde{L}}(x) &= 1 && 40 \le x \le 55 \\ \mu_{\tilde{L}}(x) &= R((x - 55)/15) && x \ge 55 \end{aligned} \tag{30}$$

For fuzzy linguistic attributes, the spreads p and q are assigned by domain experts. For fuzzy number and interval attributes, the values of p and q are considered to be $c\underline{m}$ and $c\bar{m}$, respectively. The default value of c is 0.1, but it can be changed by users. For example, for the attribute "pressure" with fuzzy interval value of "about 100–150," the default values of p and q are 10 and 15.

The section above, Other Similarity Measures, discussed several similarity functions and their main properties. It can be seen from Table 1 that $M5_{A,B}$ does not always hold the property $\mathbf{A} = \mathbf{B} \Leftrightarrow M_{A,B} = 1$. $M5_{A,B} = 1$ when $\mathbf{a} = (0.5, 1, 0.5)$ and $\mathbf{b} = (0.5, 1, 0.5)$. However, when $\mathbf{a} = (0.5, 0.5, 0.5)$ and $\mathbf{b} = (0.5, 0.5, 0.5)$, $M5_{A,B} = 0.5 \neq 1$. $M2_{A,B}$ and $M4_{A,B}$ do not always hold the property $\mathbf{A} \cap \mathbf{B} = 0 \Leftrightarrow M_{A,B} = 0$. For instance, if $\mathbf{A} = 1$ and $\mathbf{B} = 0$, then $\mathbf{A} \cap \mathbf{B} = 0$ and $M2_{A,B} = M4_{A,B} = 0$. But when $\mathbf{a} = (0.8, 0, 0)$ and $\mathbf{b} = (0, 0.9, 0)$, then $\mathbf{A} \cap \mathbf{B} = 0$, $M4_{A,B} = 1 - (0.8 + 0.9 + 0)/3 = 0.43 \neq 0$. $M2_{A,B} = 1 - \max(0.8, 0.9, 0) = 0.1 \neq 0$. So these similarity functions are not suitable for our use either. Accuracy and simplicity are the other major considerations. Commonly, discretizing the membership function yields lower

accuracy. Our area ratio method has advantages in both accuracy and simplicity. The area ratio method calculates the similarity between two fuzzy values as the ratio of the overlapping area of two corresponding membership functions to the total area as follows:

$$\text{sim}(x_i, y_i) = A(x_i \cap y_i)/A(x_i \cup y_i) = A(x_i \cap y_i)/(A(x_i) + A(y_i) - A(x_i \cap y_i)) \tag{31}$$

where A represents the area of the corresponding membership function and $x_i \cap y_i$ is the intersection of two fuzzy sets. For fuzzy number and interval attributes, five types of intersection can be classified as depicted in Figure 3. For fuzzy linguistic attributes, only types (a), (b), and (c) apply. The following algorithm (FSM) is developed to calculate the fuzzy similarity $\text{sim}(x_i, y_i)$ for these types.

Algorithm FSM // Similarity measure for two fuzzy sets x_i and y_i.

(1) $c_{xi} \leftarrow (\underline{m}_{xi} + \bar{m}_{xi})/2$, $c_{yi} \leftarrow (\underline{m}_{yi} + \bar{m}_{yi})/2$ // calculate centers of two fuzzy sets.
(2) if $c_{xi} > c_{yi}$, then exchange x_i and y_i, endif
(3) $x_i^* \leftarrow (q_{xi}\underline{m}_{yi} + p_{yi}\bar{m}_{xi})/(p_{yi} + q_{xi})$, $y_i^* \leftarrow 1 - (x_i^* - \bar{m}_{xi})/q_{xi}$ // calculate intersection point (x_i^*, y_i^*).
(4) if $y_i^* \le 0$, then $A(x_i \cap y_i) \leftarrow 0$, $\text{sim}(x_i, y_i) \leftarrow 0$ // it belongs to type (a).
(5) else $A(x_i) \leftarrow (2\bar{m}_{xi} + q_{xi} - 2\underline{m}_{xi} + p_{xi})/2$, $A(y_i) \leftarrow (2\bar{m}_{yi} + q_{yi} - 2\underline{m}_{yi} + p_{yi})/2$ // calculate $A(x_i)$, $A(y_i)$.
(6) if $0 < y_i^* < 1$, then $A(x_i \cap y_i) \leftarrow ((\bar{m}_{xi} + q_{xi} - \underline{m}_{yi} + p_{yi})y_i^*)/2$ // it belongs to type (b).
(7) else // it belongs to type (c), (d), or (e).
(8) if $\bar{m}_{xi} + q_{xi} < \bar{m}_{yi} + q_{yi}$ and $\underline{m}_{xi} - p_{xi} < \underline{m}_{yi} - p_{yi}$, then // it belongs to type (c).
(9) $A(x_i \cap y_i) \leftarrow (2\bar{m}_{xi} + q_{xi} - 2\underline{m}_{yi} + p_{yi})/2$

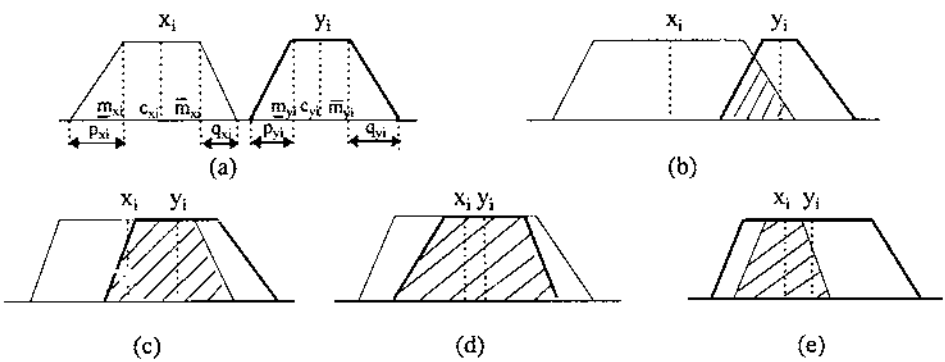


FIGURE 3. Different similarity types of two fuzzy sets x_i and y_i.

(10) else $A(x_i \cap y_i) \leftarrow \min(A(x_i), A(y_i))$ // it belongs to type (d) or (e).
(11) endif
(12) endif
(13) $\text{sim}(x_i, y_i) \leftarrow A(x_i \cap y_i)/(A(x_i) + A(y_i) - A(x_i \cap y_i))$
(14) endif
(15) end_of_FSM.

In our study, the similarities of two fuzzy linguistic attributes for all possible combinations of fuzzy sets are calculated and stored in the system in advance. When matching cases, these values are retrieved from the database if needed to save computing time. This is sensible because all fuzzy sets are predefined for each fuzzy linguistic attribute. For the similarity between two fuzzy number or interval attributes, the distance between centers c_{xi} and c_{yi} is also considered, and the similarity is computed when two cases are matched. The weighted average operator is used to aggregate the similarities of all attributes.

In summary, a hybrid similarity measure proposed for matching two cases **X** and **Y** is

$$\text{SIM}(\mathbf{X}, \mathbf{Y}) = \sum_i w_i(\varepsilon_{i_1} \text{ sim1}_{\text{FNI}}(x_i, y_i) + \varepsilon_{i_2} \text{ sim2}_{\text{FNI}}(x_i, y_i)) + \sum_j w_j \text{ sim}_{\text{FL}}(x_j, y_j) + \sum_k w_k \text{ sim}_{\text{CS}}(x_k, y_k) + \sum_l w_l \text{ sim}_{\text{CN}}(x_l, y_l) \quad (32)$$

where Eq. (31) is used to calculate $\text{sim1}_{\text{FNI}}(x_i, y_i)$ and $\text{sim}_{\text{FL}}(x_j, y_j)$, and

$$\text{sim2}_{\text{FNI}}(x_i, y_i) = 1 - \text{dist}(c_{xi}, c_{yi}) = 1 - |c_{xi} - c_{yi}|/|\max_i - \min_i|$$
$$\text{sim}_{\text{CS}}(x_k, y_k) = 1 \qquad x_k = y_k$$
$$\text{sim}_{\text{CS}}(x_k, y_k) = 0 \qquad \text{otherwise}$$
$$\text{sim}_{\text{CN}}(x_l, y_l) = 1 - \text{dist}(x_l, y_l) = 1 - |x_l - y_l|/|\max_l - \min_l|$$

where w_i, w_j, w_k, and w_l are the weights for each attribute and $\Sigma w_i + \Sigma w_j + \Sigma w_k + \Sigma w_l = 1$; $i = 1, \ldots, n1$; $j = n1 + 1, \ldots, n1 + n2$; $k = n1 + n2 + 1, \ldots, n1 + n2 + n3$; $l = n1 + n2 + n3 + 1, \ldots n$; $n1$, $n2$, $n3$, and n are the number of fuzzy number and interval, fuzzy linguistic, crisp symbolic, and total attributes, respectively. The default values of ε_{i_1} and ε_{i_2} are taken as follows:

$$\varepsilon_{i_1} = \min\left[\max_{v_{ij}, j=1,\ldots,t}(\max(v_{ij}) - \min(v_{ij}))\Big/\left(\max_{v_{ij}, j=1,\ldots,t} - \min_{v_{ij}, j=1,\ldots,t}(v_{ij})\right), 0.9\right]$$

$$\varepsilon_{i_2} = 1 - \varepsilon_{i_1}$$

where v_{ij} denotes the value of the ith attribute, and t is the total number of values having occurred in the ith attribute, which equals to the number of cases in the case base plus 1. The related parameters such as ε_{i_1}, $\max(v_{ij})$, and $\min(v_{ij})$ are also saved in the case base as facets of the ith attribute and can be updated dynamically as new cases are added. Note that the minimum value of ε_{i_2} is 0.1 to assure that the similarity between two values without intersection can be measured. In Eq. (32), the higher the weight, the more important the attribute is. In our study the weights are assigned by experts using Saaty's eigenvector method.

The matching score for the attributes with missing values is estimated as follows. In the initial analysis stages, the attributes with missing values are ignored so that more similar cases can be retrieved. They are then used to identify the most important missing data (attribute). The most important missing data could be found on the basis of an information theoretical criterion or simply the most common one shared by the retrieved cases. The identification of this attribute provides guidance for the incremental failure analysis process. In the last analysis stage, for symbolic and linguistic attributes, the matching score for missing attributes is estimated to be $1/N_i$ regardless if both or one value is unknown (N_i is the number of possible values of the ith attributes). For crisp numeric, fuzzy number, and interval attributes, the matching score is estimated to be 0.5 regardless if both or one value is unknown.

Dominant attributes can be considered as hard constraints to filter out unsuitable cases or placed at higher level nodes in the hierarchical indexing tree.

Test Example

The following example shows how the proposed hybrid measure works. Two cases **X**(case1) and **Y**(case2) of failure analysis are used; $n1$, $n2$, $n3$, and n are 2, 3, 7, and 16, respectively. The type and weight for each attribute are listed in Table 2, where CS, CN, FL, FN, and FI denote crisp symbolic, crisp numeric, fuzzy linguistic, fuzzy number, and fuzzy interval types, respectively. For the CS attributes, taking "form" as an example, the weighted similarity for this attribute is $0.1 \times 1 = 0.1$ because the values of **X** and **Y** both are "pipe" and the weight is 0.1. For the attribute "flow rate," its type is FN, x_i is about 115, and y_i is about 80. Applying the FSM algorithm, $c_{xi} = 115$ and $c_{yi} = 80$. Because of $c_{xi} > c_{yi}$, exchange x_i and y_i. Thus $c_{xi} = 80$ and $c_{yi} = 115$. Also,

$$\underline{m}_{xi} = \bar{m}_{xi} = 80 \qquad \underline{m}_{yi} = \bar{m}_{yi} = 115 \qquad p_{xi} = q_{xi} = 8 \qquad p_{yi} = q_{yi} = 11.5$$

$$x_i^* = (8 \times 115 \times 11.5 \times 80)/11.5 + 8) = 94.3590$$

$$y_i^* = 1 - (94.3590 - 80)/8 = -0.7949$$

Table 2 Example with Hybrid Similarity Measure

Attribute	Type	W_i	X	Y	Similarity
ID	CS	0	00V1143	00V1185	0
Form	CS	0.1	pipe	pipe	0.1
Crack size	FL	0.08	small	large	0.0026
Crack number	FL	0.07	many	medium	0.0052
Temperature change	FL	0.06	large	medium	0.0041
Pressure	FI	0.06	100–150 kPa	130–200 kPa	0.0413
Material	CS	0.1	carbon steel	carbon steel	0.1
Service time	CN	0.01	5.5Y	4Y	0.0097
Chemical condition	CS	0.1	acid	acid	0.1
Surface debris	CS	0.09	rust	rust	0.09
Symptom	CS	0.1	leaking	leaking	0.1
Flow rate	FN	0.02	115 mm/s	80 mm/s	0.0112
Hardness	CN	0.05	159 HK	197 HK	0.0481
Tensile strength	CN	0.05	519 MPa	508 MPa	0.0495
Fracture toughness K_{IC}	CN	0.04	60 MPa m	66 Mpa m	0.0395
Service type	CS	0.07	continuous	continuous	0.07
Total		1			0.7712

So it belongs to type (a) in Figure 3 and $\text{sim1}_{\text{FNI}}(x_i, y_i) = 0$, $\text{sim2}_{\text{FNI}}(x_i, y_i) = 1 - |80 - 115|/(500 - 0) = 1 - 0.07 = 0.93$. Say retrieved ε_{i_1} is 0.4, which is not modified by the new case, so $\varepsilon_{i_2} = 1 - 0.4 = 0.6$. Therefore the weighted similarity is $0.02 \times (0.4 \times 0 + 0.6 \times 0.93) = 0.0112$.

Taking "pressure" as an example of the FI type attribute, x_i is "about 100–150" and y_i is "about 130–200." According to the algorithm, c_{xi} and c_{yi} are 125 and 165, respectively, and

$$\underline{m}_{xi} = 100 \quad \bar{m}_{xi} = 150 \quad \underline{m}_{yi} = 130 \quad \bar{m}_{yi} = 200$$

$$p_{xi} = 10 \quad q_{xi} = 15 \quad p_{yi} = 13 \quad q_{yi} = 20$$

$$x_i = (15 \times 130 + 13 \times 150)/(13 + 15) = 139.2857$$

$$y_i^* = 1 - (139.2857 - 150)/15 = 1.7143$$

$$A(x_i) = (2\bar{m}_{xi} + q_{xi} - 2\underline{m}_{xi} + p_{xi})/2 = (2 \times 150 + 15 - 2 \times 100 + 10)/2 = 62.5$$

$$A(y_i) = (2\bar{m}_{yi} + q_{yi} - 2\underline{m}_{yi} + p_{yi})/2 = (2 \times 200 + 20 - 2 \times 130 + 13)/2 = 86.5$$

Because

$$\bar{m}_{xi} + q_{xi} = 150 + 15 = 165 < \bar{m}_{yi} + q_{yi} = 200 + 20 = 220$$

$$\underline{m}_{xi} - p_{xi} = 100 - 10 = 90 < \underline{m}_{yi} - p_{yi} = 130 - 13 = 117$$

it belongs to type (c) and

$$A(x_i \cap y_i) = (2\bar{m}_{xi} + q_{xi} - 2\underline{m}_{yi} + p_{yi})/2$$
$$= (2 \times 150 + 15 - 2 \times 130 + 13)/2 = 34$$
$$\text{sim1}_{\text{FNI}}(x_i, y_i) = A(x_i \cap y_i)/(A(x_i) + A(y_i) - A(x_i \cap y_i))$$
$$= 34/(62.5 + 86.5 - 34) = 0.2957$$
$$\text{sim2}_{\text{FNI}}(x_i, y_i) = 1 - |125 - 165|/(320 - 0) = 1 - 0.125 = 0.875$$

Say retrieved ε_{i_1} is 0.25. Because of $(200 + 20) - (130 - 13) = 103$ and $103/320 = 0.3219 > 0.25$, ε_{i_1} is replaced with 0.3219 and $\varepsilon_{i_2} = 1 - 0.3219 = 0.6781$. So the weighted similarity for this attribute is $0.06 \times (0.3219 \times 0.2957 + 0.6781 \times 0.875) = 0.0413$.

As an example of the FL attributes, the values of "extent of temperature change" in **X** and **Y** are "large" and "medium." The similarity value of "large" and "medium" is 0.0685, which is retrieved from the database. The 0.0685 value was precalculated according to the FSM algorithm based on the membership functions given in Eqs. (29) and (30). Therefore the weighted similarity for this attribute is $0.06 \times 0.0685 = 0.0041$. An example of CN attributes is "service time," the values of the attribute in **X** and **Y** are "5.5" and "4," the corresponding weighted similarity is $0.01 \times (1 - |5.5 - 4|/(50 - 0)) = 0.01 \times 0.97 = 0.0097$. Thus the total similarity of **X** and **Y**, according to Eq. (32), is 0.7712.

Table 2 demonstrates that, without the fuzzy measure, the similarity for three FL types attributes cannot be measured and the related similarities $0.0026 + 0.0052 + 0.0041 = 0.0119$ will be lost. For the FI and FN type attributes, only the distances between corresponding centers can be used to calculate the similarities. It is clear that the total similarity with a pure crisp similarity measure for the same cases, **X** and **Y**, cannot reflect the real similarity correctly.

CONCLUSION

We have reviewed various similarity measures in the literature and proposed a hybrid similarity measure for handling applications with a mixture of crisp and fuzzy features. Its application to failure analysis is illustrated. Results show that our hybrid measure can reflect the domain nature effectively. The more the fuzziness in the case attributes, the more the power of

the proposed measure. It is thus shown that combining fuzzy set theory with the conventional CBR system greatly enhances its capability.

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